IE540
L9. Policy based RL : Stochastic PG

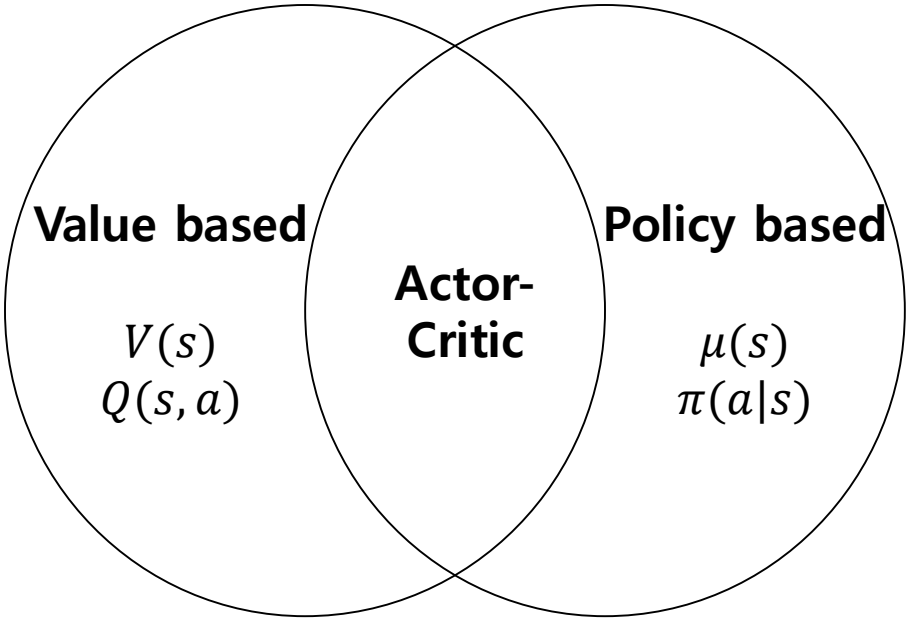
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Overview

- Model is known → DP (in Lecture 3)
  - Evaluation (prediction) : Iterative policy evaluation
  - Control (finding optimal solution)
- Model is unknown → RL (by use of interaction samples)
  - Value based RL
    - MC method : Evaluation, Control (in Lecture 4)
    - TD method : Q, SARSA (in Lecture 5)
    - n-step bootstrapping (in Lecture 6)
    - value function approximation (in Lecture 7)
    - NN, DQN (Lecture 8)
  - Policy based RL
    - Stochastic Policy Gradient
      - PG Intro, REINFORCE, Actor-Critic (in Lecture 9)
      - TRPO, PPO (in Lecture 10)
    - Deterministic Policy Gradient, Derivative free methods (in Lecture 11)
- Exploration : MAB

Value-based RL vs Policy-based RL

- Value-based
  - Learn value function
    - $V(s)$ or $Q(s,a)$
  - Implicit policy
    - $\mu(s)$ = argmax $Q(s,a)$
    - ε-greedy
- Policy-based
  - No value function
  - Learn policy directly & explicitly
    - $\mu(s)$: deterministic policy
    - $\pi(a|s)$: stochastic policy
- Actor-Critic
  - Learn both
    - Actor : policy $\mu(s)$ or $\pi(a|s)$
    - Critic : value function $V(s)$ or $Q(s,a)$



Why policy-based RL?

- Advantages
  - Policy function can be much simpler than value function
  - Effective in high-dim. or continuous action space
    - Value based method needs to find $\underset{a}{\operatorname{argmax}}\, Q(s,a)$: hard for cont. action
  - Can learn stochastic policy
  - Policy : direct goal, cf. value function : indirect intermediary
- Disadvantages
  - Local optimal convergence
  - High variance in policy evaluation
    - value based : $\max_a Q(s,a)$ is a very aggressive update operator
      - bad estimate on some $Q(s,a)$ → only limited harmful effect
    - policy based opt. (policy gradient) usually can take only small steps
      - small step → slow update
      - large step → bad policy → useless trajectory obtained → can't recover
  - Low sample efficiency : how to reuse earlier samples?
    - cf. Experience replay in DQN

Policy based RL (Policy Optimization)

- **Policy Gradient**
  - Stochastic policy gradient
    - REINFORCE
    - Actor-Critic : Stochastic policy (actor) & Value function (critic)
      - Q A-C : Critic is Q(s,a)
      - Advantage A-C (A2C), Async. Adv. A-C (A3C, Deepmind, ICML2016)
      - Other variants : TD A-C, TD(λ) A-C, …
    - TRPO(ICML 2015), PPO(arXiv 2017), PPG(arXiv 2020) : by J. Schulman
    - SAC (Haarnoja, ICML 2018)
  - Deterministic policy gradient
    - DPG(Silver, ICML 2014)
    - DDPG(Deepmind, ICLR 2016)
    - TD3 (Fujimoto, ICML 2018)
- **Derivative Free Optimization**
  - Evolutionary strategy (ES) : population based
    - CEM (Cross-Entropy Method)
    - GA (Genetic Algorithm), SA (Simulated Annealing)
- **PG + ES** : Combine PG with ES
  - ERL (NIPS 2018) : GA + DDPG
  - CEM-RL (ICLR 2019) : CEM + TD3
  - …

Some Definitions

- **MDP:** $< \mathcal{S}, \mathcal{A}, P, r, p_0, \gamma >$
  - ✓ $\mathcal{S}$: state set, $\mathcal{A}$: action set
  - ✓ $p(s'|s,a)$: state transition probability distribution,
  - ✓ $r(s,a)$: reward function, expected reward for action a at state s
  - ✓ $p_0(s_0)$: distribution of the initial state s_0
  - ✓ $\gamma \in (0,1]$: discount factor
- **Policy :** parameterized policy function
  - ✓ $\pi_\theta(a|s)$: stochastic policy, probability dist. of actions at state s
  - ✓ $\mu_\theta(s)$: deterministic policy $\mu: S \to A$
- **Trajectory** $\tau = (s_0, a_0, r_0, s_1, a_1, r_1, s_2, \dots, r_{T-1}, s_T)$
  - ✓ Partial trajectory : $\tau_t = (s_t, a_t, r_t, s_{t+1}, a_{t+1}, r_{t+1}, s_{t+2}, \dots, r_{T-1}, s_T)$
  - ✓ $\tau \sim \pi$: trajectory where actions are chosen by π
  - ✓ $P(\tau;\pi) = p_0(s_0) \prod_{t=0}^{T-1} \pi(a_t|s_t) p(s_{t+1}|s_t, a_t)$
- **Value functions**
  - $V^{\pi_\theta}(s) = \mathbb{E}[G|\pi_\theta] = \mathbb{E}[r_0 + \gamma r_1 + \gamma^2 r_2 + \cdots |\pi_\theta]$
  - $V^{\pi_\theta}(s) = \mathbb{E}_{a\sim\pi_\theta}[Q^{\pi_\theta}(s,a)]$

Objective functions for policy optimization

$$d^{\pi_\theta}(s) = \frac{\rho^{\pi_\theta}(s)}{\sum_{x \in \mathcal{S}} \rho^{\pi_\theta}(x)}$$

$$\rho^{\pi_\theta}(s) = \sum_{s_0} p_0(s_0) \sum_{k=0}^{\infty} \gamma^k \, p(s_0 \to s, k, \pi_\theta)$$

- policy π_θ : parameterized stochastic policy
- Objective function $J(\theta)$: Policy opt. is to find $\theta^* = \underset{\theta}{\operatorname{argmax}} J(\theta)$
  - Many forms of objective function → similar result (see A1)
  - Expected return of an episode (for episodic env.) : also called "start value"
    - Expected return (value function view)

$$J_0(\theta) = \mathbb{E}_{s_0 \sim p_0}[V^{\pi_\theta}(s_0)] = \sum_{s_0 \in \mathcal{S}} p_0(s_0) V^{\pi_\theta}(s_0), \qquad p_0(s_0): \text{initial state dist.}$$

$$= \mathbb{E}_{s_0 \sim p_0}\left[\mathbb{E}_{a \sim \pi_\theta}[Q^{\pi_\theta}(s_0, a)]\right] = \sum_{s_0 \in \mathcal{S}} p_0(s_0) \sum_{a \in \mathcal{A}} \pi_\theta(a|s_0) Q^{\pi_\theta}(s_0, a)$$

    - Expected return **(trajectory return view)** : identical to value function view

$$J_0(\theta) = \mathbb{E}_{\tau \sim p_0, \pi_\theta}[R(\tau)], \text{where } R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t \text{ : we will use this mainly}$$

  - Expected value w.r.t. stationary dist. (continuing env.)

$$J_{EV}(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta}}[V^{\pi_\theta}(s)] = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta}[Q^{\pi_\theta}(s, a)] = \sum_{s \in \mathcal{S}} d^{\pi_\theta}(s) \sum_{a \in \mathcal{A}} \pi_\theta(a|s) Q^{\pi_\theta}(s, a)$$

  - Expected reward per step (continuing env.)

$$J_{ER}(\theta) = \mathbb{E}_{s \sim d^{\pi_\theta},\, a \sim \pi_\theta}[r(s, a)] = \sum_{s \in \mathcal{S}} d^{\pi_\theta}(s) \sum_{a \in \mathcal{A}} \pi_\theta(a|s) r(s, a)$$

  - for continuing task : $J_0(\theta) \propto J_{EV}(\theta) = \frac{1}{1-\gamma} J_{ER}(\theta)$

(Stochastic) Policy Gradient Theorem

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)] \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P\left(\tau^{(i)};\theta\right) R\left(\tau^{(i)}\right)$$

[proof]

$$J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[R(\tau)] = \sum_\tau P(\tau;\theta)R(\tau)$$

$$\nabla_\theta J(\theta) = \nabla_\theta \sum_\tau P(\tau;\theta)R(\tau) = \sum_\tau \nabla_\theta P(\tau;\theta)R(\tau)$$

$$= \sum_\tau P(\tau;\theta)\frac{\nabla_\theta P(\tau;\theta)}{P(\tau;\theta)}R(\tau) = \sum_\tau P(\tau;\theta)\nabla_\theta \log P(\tau;\theta)\, R(\tau)$$

$$= \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)]$$

Interpretation : make high return trajectory $(R(\tau))$ more probable $(P(\tau;\theta))$

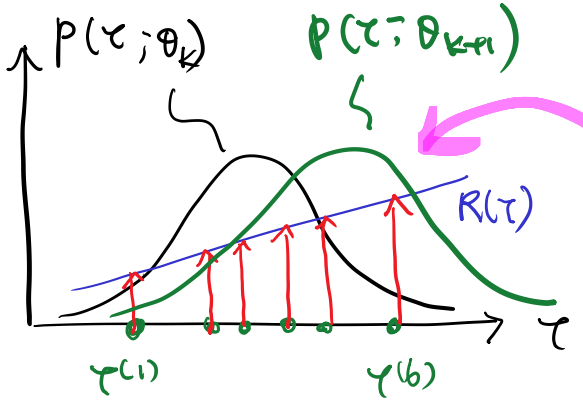
$$R(\tau) = \sum_{t=0}^{T-1} \gamma^t r_t$$

$$P(\tau;\theta) = p_0(s_0)\prod_{t=0}^{T-1} \pi_\theta(a_t|s_t)p(s_{t+1}|s_t,a_t)$$

$$\nabla_{\theta=\theta_k} f(\theta) \doteq \frac{d}{d\theta} f(\theta)\Big|_{\theta=\theta_k}$$

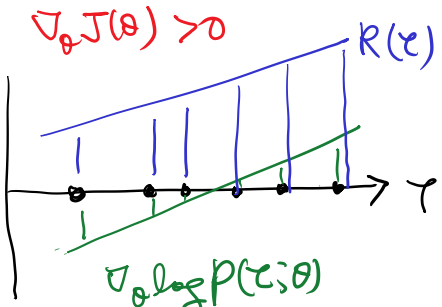
$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)] \approx \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P\left(\tau^{(i)};\theta\right) R\left(\tau^{(i)}\right)$$

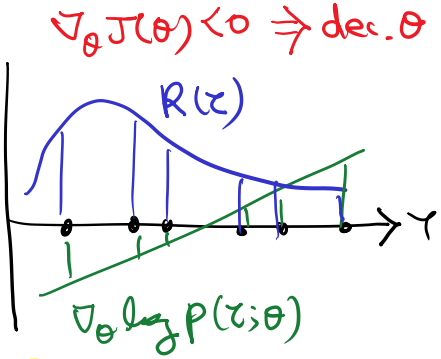
$$\theta_{k+1} \leftarrow \theta_k + \alpha \nabla_{\theta=\theta_k} J(\theta) = \theta_k + \frac{\alpha}{N}\sum_{i=1}^{N} \nabla_{\theta=\theta_k} \log P(\tau^{(i)};\theta) \cdot R(\tau^{(i)})$$



inc. $P(\tau^{(i)};\theta)$ if $R(\tau^{(i)})$ > average return
dec " otherwise

Ex. $P(\tau;\theta) \propto \exp(-(\tau-\theta)^2)$ $\tau \sim N(\theta,1)$
$\nabla_\theta \log P(\tau;\theta) \propto (\tau-\theta)$





$$\mathbb{E}_{\tau\sim P(\tau;\theta)}[\nabla_\theta \log P(\tau;\theta)]$$

$$= \sum_\tau P(\tau;\theta)\frac{\nabla_\theta P(\tau;\theta)}{P(\tau;\theta)}$$

$$= \sum_\tau \nabla_\theta P(\tau;\theta) = \nabla_\theta\left(\sum_\tau P(\tau;\theta)\right) = \nabla_\theta(1) = 0 \Rightarrow \frac{1}{N}\sum_{i=1}^{N} \nabla_\theta \log P(\tau^{(i)};\theta) \approx 0$$

(Stochastic) Policy Gradient Theorem

$$J(\theta) = \mathbb{E}_{\tau \sim p_0, \pi_\theta}[R(\tau)] = \sum_{\tau} P(\tau;\theta) R(\tau), \qquad P(\tau;\theta) = p_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t)$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_0, \pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)] = \mathbb{E}_{\tau \sim p_0, \pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, R(\tau)\right]$$

PG w/ trajectory

PG w/ policy π_θ

$$\nabla_\theta \log P(\tau;\theta) = \nabla_\theta \log \left[p_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t|s_t) p(s_{t+1}|s_t, a_t) \right]$$

$$= \nabla_\theta \left[\log p_0(s_0) + \sum_{t=0}^{T-1} \log \pi_\theta(a_t|s_t) + \sum_{t=0}^{T-1} \log p(s_{t+1}|s_t, a_t) \right] = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)$$

$R(\tau) = \left(\sum_{t=0}^{T-1} \gamma^t r_t \right)$: return of τ

$\left(\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \right)$: sum of scores

TAQ. Why not deterministic policy?

Derivation of PG using importance sampling

$$J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[R(\tau)] = \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\frac{P(\tau;\theta)}{P(\tau;\theta_k)}R(\tau)\right] \quad \pi_\theta \neq \pi_{\theta_k}$$ (importance sampling)

$$\nabla_\theta J(\theta) = \nabla_\theta \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\frac{P(\tau;\theta)}{P(\tau;\theta_k)}R(\tau)\right] = \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\frac{\nabla_\theta P(\tau;\theta)}{P(\tau;\theta_k)}R(\tau)\right]$$

$J(\theta) = \theta^2$
$\nabla_\theta J(\theta) = 2\theta$
$\nabla_{\theta=\theta_k} J(\theta) = 2\theta_k$

Let's denote $\nabla_\theta J(\theta)\big|_{\theta=\theta_k} = \nabla_{\theta=\theta_k} J(\theta)$

$$\nabla_{\theta=\theta_k} J(\theta) = \frac{dJ(\theta)}{dx}\bigg|_{\theta=\theta_k}$$

$$\nabla_{\theta=\theta_k} J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\frac{\nabla_{\theta=\theta_k} P(\tau;\theta)}{P(\tau;\theta_k)}R(\tau)\right] = \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\nabla_{\theta=\theta_k}\log P(\tau;\theta)\,R(\tau)\right]$$

$$= \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}\left[\sum_{t=0}^{T-1}\nabla_{\theta=\theta_k}\log\pi_\theta(a_t|s_t)\,R(\tau)\right]$$

Interpretation

$$\max_\theta \mathbb{E}_{\tau\sim p_0,\pi_\theta}[R(\tau)] \equiv \max_\theta \mathbb{E}_{\tau\sim p_0,\pi_{\theta_k}}[\log P(\tau;\theta)\,R(\tau)]$$

$$\approx \max_\theta \frac{1}{N}\sum_{i=1}^{N}\log P\left(\tau^{(i)};\theta\right)R\left(\tau^{(i)}\right), \text{ where } \tau^{(i)} \text{ is sampled under } \pi_{\theta_k}$$

→ Increase probability for higher return trajectory $\tau^{(i)}$

How to compute PG?

Let's replace $\mathbb{E}_\tau$ with a sample trajectory $\tau = (s_0, a_0, r_0, \dots, r_{T-1}, s_T)$

$$\nabla_\theta J(\theta) \approx \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, R(\tau)\,, \text{where } R(\tau) = \sum_{k=0}^{T-1} \gamma^k r_k$$

Here, evaluating the choice of a_t using $R(\tau)$ doesn't make sense.

→ $R(\tau)$ includes earlier rewards $(r_0, r_1, \dots, r_{t-1})$ prior to a_t

→ Better to use the return of the partial trajectory $\tau_t = (s_t, a_t, r_t, \dots, r_{T-1}, s_T)$

'Return from t' : $R_t = R(\tau_t) = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$ (see A2)

$$\nabla_\theta J(\theta) \approx \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, R_t, \text{ where } R_t = R(\tau_t) = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$$

g_t: gradient at time step t

REINFORCE Algorithm (aka Vanilla PG)

➢ **Algorithm REINFORCE** (per trajectory update)

1. initialize policy $\pi_\theta(a|s)$
2. for e = 1…N (# of episodes)
   - generate an episode $\tau = (s_0, a_0, r_0, s_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}, s_T)$ following $\pi_\theta(a|s)$
     - also save $L_t = \log \pi_\theta(a_t|s_t)$ at each step t
   - compute return $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$ at each step t
     - $R_T = 0$; $R_t = r_t + \gamma R_{t+1}$ for $t = T-1, \dots, 1, 0$
   - update policy : $\theta \leftarrow \theta + \alpha \nabla_\theta \sum_{t=0}^{T-1} L_t R_t$

$$\min_\theta \; loss = -\sum_{t=0}^{T-1} L_t R_t$$

➢ **Algorithm REINFORCE** (per step update)

1. initialize policy $\pi_\theta(a|s)$
2. for e = 1…N (# of episodes)
   - generate an episode $\tau = (s_0, a_0, r_0, s_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}, s_T)$ following $\pi_\theta(a|s)$
     - also save $L_t = \log \pi_\theta(a_t|s_t)$ at each step t
   - $R_T = 0$
   - for $t = T-1, \dots, 1, 0$
     - compute return $R_t = r_t + \gamma R_{t+1}$
     - update policy : $\theta \leftarrow \theta + \alpha \nabla_\theta L_t R_t$

TAQ. Is this more efficient?

REINFORCE Algorithm

- MC update, not TD update (Update after an episode is completed)
- Computing return : $R_t = r_t + \gamma R_{t+1},\ t = T-1, \dots, 0$
  - If episode τ has ended at time T: $R_T = 0$
  - If not (trajectory stopped before ending), $R_T = V^{\pi_\theta}(s_T)$

TAQ. Do we have $V^{\pi_\theta}(s)$?

- Variance of R_t can be high → how to reduce variance of $\nabla_\theta J(\theta)$?

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\, R(\tau)] = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\,(R(\tau) - b)]$$

  - Good choice for baseline b ?
    - $b = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[R(\tau)] \approx \frac{1}{n}\sum_{i=1}^{n} R(\tau^{(i)})$
    - $b = \frac{\sum_{i=1}^{n}\left(\nabla_\theta \log P(\tau^{(i)};\theta)\right)^2 R(\tau^{(i)})}{\sum_{i=1}^{n}\left(\nabla_\theta \log P(\tau^{(i)};\theta)\right)^2}$ (Greensmith et al, JMLR 2004)

$b \cdot \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)] = 0$
see A3 for proof

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_0,\pi_\theta}\left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t)\, \gamma^t (R(\tau_t) - b(s_t))\right]$$

(Computing stepwise PG)

- better choice? use the current estimate of

REINFORCE Algorithm w/ baseline

➢ **Algorithm REINFORCE** (w/ baseline)

1. initialize policy $\pi_\theta(a|s)$ and value $V_v(s_t)$
2. for e = 1…N (# of episodes)
   - generate an episode $\tau = (s_0, a_0, r_0, s_1, \dots, s_{T-1}, a_{T-1}, r_{T-1}, s_T)$ following $\pi_\theta(a|s)$
     - also save $L_t = \log \pi_\theta(a_t|s_t)$ at each step t
   - Compute final return : $R_T = 0$ if s_T is terminal else $V_v(s_T)$
   - compute return: $R_t = r_t + \gamma R_{t+1}$ for $t = T-1, \dots, 1, 0$
   - update policy by minimizing loss : $-\sum_{t=0}^{T-1} L_t \big(R_t - V_v(s_t)\big)$
     - $\theta \leftarrow \theta + \alpha \nabla_\theta \sum_{t=0}^{T-1} L_t \, (R_t - V_v(s_t))$
   - update $V_v(.)$ by minimizing MSE loss : $\frac{1}{T}\sum_{t=0}^{T-1} \|R_t - V_v(s_t)\|^2$
     - $v \leftarrow v + \beta \frac{2}{T} \sum_{t=0}^{T-1} \big(R_t - V_v(s_t)\big) \nabla_\theta V_v(s_t)$

➢ This can lead us to actor-critic algorithm
- a little modification of REINFORCE
- per-step update
- introduce TD(temporal difference) structure

Going further on PG

$$d^{\pi_\theta}(s) = \frac{\rho^{\pi_\theta}(s)}{\sum_{x \in \mathcal{S}} \rho^{\pi_\theta}(x)}$$

$$\rho^{\pi_\theta}(s) = \sum_{s_0} p_0(s_0) \sum_{k=0}^{\infty} \gamma^k \, p(s_0 \to s, k, \pi_\theta)$$

- Introducing per step view of PG
  - Trajectory view: $\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_0, \pi_\theta}[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(a_t|s_t) \, R_t], R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$
  - Per step view: $\nabla_\theta J(\theta) \propto \mathbb{E}_{s_t \sim d^{\pi_\theta}, a_t \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t) \, Q^{\pi_\theta}(s_t, a_t)]$
    - (see A4 or RLAI 2nd ed. p.327 for full proof)
    - allow per-step update (instead of per-episode update)
    - need to estimate $Q^{\pi_\theta}(s_t, a_t) = \mathbb{E}[R_t|s_t, a_t]$, where $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$

- State dependent baseline : $Q^{\pi_\theta}(s_t, a_t) - b(s_t)$
  - $\nabla_\theta J(\theta) \propto \mathbb{E}_{s_t \sim d^{\pi_\theta}, a_t \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t) \{Q^{\pi_\theta}(s_t, a_t) - b(s_t)\}]$
    - Unbiased estimator of PG
  - $b(s_t)$: state dependent baseline
    - Use $V^{\pi_\theta}(s_t)$ for baseline $b(s_t)$
    - $A^{\pi_\theta}(s_t, a_t) = Q^{\pi_\theta}(s_t, a_t) - V^{\pi_\theta}(s_t)$: advantage function
      - remove the effect the state s_t from $Q^{\pi_\theta}(s_t, a_t)$
      - captures the effect of the action a_t only

$b(s_t) \cdot \mathbb{E}_{a_t \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)] = 0$
see A4 for proof

Actor-Critic algorithm

- Policy Gradient revisited
  - $\nabla_\theta J(\theta) \propto \mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta}[\nabla_\theta \ln \pi_\theta(a|s)\, Q^{\pi_\theta}(s,a)]$
- REINFORCE
  - Episodic task only
  - Update after an episode completes
    - Use return G_t instead of $Q^{\pi_\theta}(s,a)$
    - computing return requires a complete episode $G_T = 0$, $G_t = R_{t+1} + \gamma G_{t+1}$
  - High variance of G_t
- Actor-Critic → Update at every step
  - Critic $Q_w(s,a)$
    - Learn value function using SARSA or Q-learning
  - Actor $\pi_\theta(a|s)$
    - Learn policy function using PG w/$Q_w(s,a)$

TAQ. How can we use NN for $\pi_\theta(a|s)$ ensuring output is probability?
i.e. $\pi_\theta(a|s) \geq 0$, $\sum_a \pi_\theta(a|s) = 1$

Vanilla A-C (Action value A-C)

➢ **Algorithm Action value Actor-Critic (Q learning form)**

1. initialize policy $\pi_\theta(.|s)$ and action value function $Q_w(s,a)$
2. for e = 1…N (# of episodes)
   - $t = 0$, sample initial state s from $p_0(s)$
   - while s is not terminal or $t < t_{\max}$
     - select $a \sim \pi_\theta(.|s)$
     - perform a, observe r, s'
     - update actor
       - $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \log \pi_\theta(a|s)\, Q_w(s,a)$
     - update critic
       - $\delta \leftarrow r + \gamma \max_{a'} Q_w(s',a') - Q_w(s,a)$
       - $w \leftarrow w - \alpha_w \delta \nabla_w Q_w(s,a)$
     - $s \leftarrow s', t \leftarrow t+1$

Advantage function

$$\nabla_\theta J(\theta) = \mathbb{E}_\tau\left[\sum_{t=0}^{T-1} R(\tau)\nabla_\theta \log \pi_\theta(a_t|s_t)\right] = \mathbb{E}_\tau\left[\sum_{t=0}^{T-1} (R(\tau_t) - b(s_t))\nabla_\theta \log \pi_\theta(a_t|s_t)\right]$$

Let $\hat{G} = \sum_{t=0}^{T-1} \phi(\tau, s_t)\nabla_\theta \log \pi_\theta(a_t|s_t)$. What can be used for $\phi(\tau, s_t)$?

- $R(\tau) = \sum_{k=0}^{T-1} \gamma^k r_k$: total return
- $R_t = \sum_{k=t}^{T-1} \gamma^{k-t} r_k$: return after action a_t smaller var. than $R(\tau)$
- $(R_t - b)$, $(R_t - V^{\pi_\theta}(s_t))$: baselined return, to reduce var. of R_t
- $Q^{\pi_\theta}(s_t, a_t)$: action value
- $A^{\pi_\theta}(s_t, a_t) = Q^{\pi_\theta}(s_t, a_t) - V^{\pi_\theta}(s_t)$: advantage function
  - smaller var., but need to estimate both of $Q^{\pi_\theta}(s_t, a_t)$ and $V^{\pi_\theta}(s_t)$
- $\delta^{\pi_\theta} = r + \gamma V^{\pi_\theta}(s') - V^{\pi_\theta}(s)$: TD residual

  $\mathbb{E}_{a\sim\pi_\theta}[\delta^{\pi_\theta}] = \mathbb{E}_{a\sim\pi_\theta}[r + \gamma V^{\pi_\theta}(s') - V^{\pi_\theta}(s)]$

  $= \mathbb{E}_{a\sim\pi_\theta}[r + \gamma V^{\pi_\theta}(s')] - V^{\pi_\theta}(s) = Q^{\pi_\theta}(s, a) - V^{\pi_\theta}(s)$
  - This bootstrapping introduces bias when $V^{\pi_\theta}(s)$ is not exactly known

TD Advantage A-C

➢ **Algorithm Advantage Actor-Critic**

1. initialize policy $\pi_\theta(.|s)$ and state value function $V_w(s)$
2. for e = 1…N (# of episodes)
   - $t = 0$, sample initial state s from $p_0(s)$
   - while s is not terminal or $t < t_{\max}$
     - select $a \sim \pi_\theta(.|s)$
     - perform a, observe r, s'
     - update actor
       - $\delta = r + \gamma V_w(s') - V_w(s)$
       - $\theta \leftarrow \theta + \alpha_\theta \delta \nabla_\theta \log \pi_\theta(a|s)$
     - update critic
       - $w \leftarrow w - \alpha_w \delta \nabla_w V_w(s)$
     - $s \leftarrow s'$, $t \leftarrow t + 1$

$$\min_\theta \; -\big(r + \gamma V_w(s') - V_w(s)\big) \log \pi_\theta(a|s)$$

$$\min_w \; \big(r + \gamma V_w(s') - V_w(s)\big)^2$$

TAQ. Can we use experience replay for A-C?

A1. $J_0(\theta) = J_{ER}(\theta)$ [proof]

$$V^{\pi_\theta}(s) = r^{\pi_\theta}(s) + \gamma \sum_a \pi_\theta(a|s) \sum_{s'} p(s'|s,a) V^{\pi_\theta}(s') \quad \because r^{\pi_\theta}(s) = \mathbb{E}_{a\sim\pi_\theta(.|s)}[r(s,a)]$$

$$= r^{\pi_\theta}(s) + \gamma \sum_{s'} p(s \to s', 1, \pi_\theta) V^{\pi_\theta}(s') \quad \because p(s \to s', 1, \pi_\theta) = \sum_a \pi_\theta(a|s) p(s'|s,a)$$

$$= r^{\pi_\theta}(s) + \gamma \sum_{s'} p(s \to s', 1, \pi_\theta) \left[r^{\pi_\theta}(s') + \gamma \sum_{s''} p(s' \to s'', 1, \pi_\theta) V^{\pi_\theta}(s'') \right]$$

$$= r^{\pi_\theta}(s) + \gamma \sum_{s'} p(s \to s', 1, \pi_\theta) r^{\pi_\theta}(s') + \gamma^2 \sum_{s''} p(s \to s'', 2, \pi_\theta) V^{\pi_\theta}(s'')$$

$$= r^{\pi_\theta}(s) + \gamma \sum_{s'} p(s \to s', 1, \pi_\theta) r^{\pi_\theta}(s') + \gamma^2 \sum_{s''} p(s \to s'', 2, \pi_\theta) r^{\pi_\theta}(s'') + \cdots$$

$$= \sum_{t=0}^{\infty} \gamma^t \sum_x p(s \to x, t, \pi_\theta) r^{\pi_\theta}(x) = \sum_x \sum_{t=0}^{\infty} \gamma^t \, p(s \to x, t, \pi_\theta) r^{\pi_\theta}(x)$$

$$J_0(\theta) = \mathbb{E}_{s_0\sim p_0}[V^{\pi_\theta}(s_0)] = \sum_{s_0} p_0(s_0) V^{\pi_\theta}(s_0) = \sum_x \sum_{s_0} \sum_{t=0}^{\infty} \gamma^t \, p_0(s_0) p(s_0 \to x, t, \pi_\theta) r^{\pi_\theta}(x)$$

$$= \sum_x \rho^{\pi_\theta}(x) r^{\pi_\theta}(x) \propto \mathbb{E}_{s\sim d^{\pi_\theta}}[r^{\pi_\theta}(s)] = \mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta}[r(s,a)] = J_{ER}(\theta)$$

$\rho^{\pi_\theta}(s) = \sum_{s_0} p_0(s_0) \sum_{k=0}^{\infty} \gamma^k \, p(s_0 \to s, k, \pi_\theta)$: discounted visitation frequency (improper distribution)

$d^{\pi_\theta}(s) = \dfrac{\rho^{\pi_\theta}(s)}{\sum_{x\in\mathcal{S}} \rho^{\pi_\theta}(x)}$: discounted stationary distribution (proper)

A2. Reward-to-go [proof]

$$\mathbb{E}_{\tau\sim p_0,\pi_\theta}[g_t] = \mathbb{E}_{a_t\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{k=0}^{T-1}\gamma^k r_k\right)\right] = \sum_{k=0}^{T-1}\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\gamma^k r_k]$$

$$= \sum_{k=0}^{t-1}\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\gamma^k r_k] + \sum_{k=t}^{T-1}\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\gamma^k r_k]$$

$$= \sum_{k=t}^{T-1}\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\gamma^k r_k] = \mathbb{E}_{a_t\sim\pi_\theta}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{k=t}^{T-1}\gamma^{k-t} r_k\right)\right]\gamma^t$$

$$\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\gamma^k r_k] = \mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)]\gamma^k r_k$$

($\because r_k$ is independent from τ_t if $k < t$)

$$\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)] = \sum_{a_t}\pi_\theta(a_t|s_t)\nabla_\theta \log \pi_\theta(a_t|s_t)$$

$$= \sum_{a_t}\pi_\theta(a_t|s_t)\frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_\theta(a_t|s_t)} = \sum_{a_t}\nabla_\theta \pi_\theta(a_t|s_t) = \nabla_\theta \sum_{a_t}\pi_\theta(a_t|s_t) = \nabla_\theta(1) = 0$$

A3. introducing baseline w/o bias [proof]

- In general, it holds for any pdf $f(x;\theta)$:

$$\mathbb{E}_{x\sim f}[\nabla_\theta \log f(x;\theta)] = \sum_x f(x;\theta)\nabla_\theta \log f(x;\theta) = \sum_x \nabla_\theta f(x;\theta) = \nabla_\theta\left(\sum_x f(x;\theta)\right) = \nabla_\theta(1) = 0$$

- Constant baseline $(R(\tau) - b)$ → unbiased estimator of policy gradient

$$\mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)] = \sum_\tau P(\tau;\theta)\nabla_\theta \log P(\tau;\theta) = \sum_\tau \nabla_\theta P(\tau;\theta) = \nabla_\theta\left(\sum_\tau P(\tau;\theta)\right) = \nabla_\theta(1) = 0$$

$$\mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\,(R(\tau) - b)] = \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\,R(\tau)] - b\,\mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)]$$
$$= \mathbb{E}_{\tau\sim p_0,\pi_\theta}[\nabla_\theta \log P(\tau;\theta)\,R(\tau)]$$

- State dependent baseline $\{Q^{\pi_\theta}(s_t,a_t) - b(s_t)\}$→ unbiased estimator

$$\mathbb{E}_{s_t\sim d^{\pi_\theta},a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,b(s_t)] = \mathbb{E}_{s_t\sim d^{\pi_\theta}}\left[\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,b(s_t)]\right]$$
$$= \mathbb{E}_{s_t\sim d^{\pi_\theta}}\left[b(s_t)\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)]\right] = \mathbb{E}_{s_t\sim d^{\pi_\theta}}\left[b(s_t)\mathbb{E}_{a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)]\right]$$
$$= \mathbb{E}_{s_t\sim d^{\pi_\theta}}[b(s_t)\cdot 0] = 0$$
$$\mathbb{E}_{s_t\sim d^{\pi_\theta},a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,\{Q^{\pi_\theta}(s_t,a_t) - b(s_t)\}] = \mathbb{E}_{s_t\sim d^{\pi_\theta},a_t\sim\pi_\theta}[\nabla_\theta \log \pi_\theta(a_t|s_t)\,Q^{\pi_\theta}(s_t,a_t)]$$

A4. per step view of $\nabla_\theta J(\theta)$ [proof]

$$J_0(\theta) = \mathbb{E}_{s\sim p_0}[V^{\pi_\theta}(s)] = \sum_{s\in\mathcal{S}} p_0(s)V^{\pi_\theta}(s) = \sum_s \sum_x \sum_{t=0}^{\infty} \gamma^t\, p_0(s)p(s\to x,t,\pi)r^{\pi}(x)$$

$$V^{\pi_\theta}(s) = \mathbb{E}_{a\sim\pi_\theta}[Q^{\pi_\theta}(s,a)] = \sum_a \pi_\theta(a|s)Q^{\pi_\theta}(s,a)\,,\; Q^{\pi_\theta}(s,a) = \mathbb{E}_{s',r\sim p(.|s,a)}[r+\gamma V^{\pi_\theta}(s')]$$

$$\nabla_\theta V^{\pi_\theta}(s) = \overbrace{\sum_a \nabla_\theta \pi_\theta(a|s)Q^{\pi_\theta}(s,a)}^{\phi(s)} + \sum_a \pi_\theta(a|s)\nabla_\theta Q^{\pi_\theta}(s,a) = \phi(s) + \sum_a \pi_\theta(a|s)\nabla_\theta Q^{\pi_\theta}(s,a)$$

$$= \phi(s) + \sum_a \pi_\theta(a|s)\nabla_\theta \left[\sum_{s',r} p(s',r|s,a)\big(r+\gamma V^{\pi_\theta}(s')\big)\right] = \phi(s) + \gamma \sum_a \pi_\theta(a|s) \sum_{s'} p(s'|s,a)\nabla_\theta V^{\pi_\theta}(s')$$

$$= \phi(s) + \gamma \sum_{s'} \nabla_\theta V^{\pi_\theta}(s') \left[\overbrace{\sum_a \pi_\theta(a|s)p(s'|s,a)}^{p(s\to s',1,\pi)}\right] = \phi(s) + \gamma \sum_{s'} p(s\to s',1,\pi)\nabla_\theta V^{\pi_\theta}(s')$$

$$= \phi(s) + \gamma \sum_{s'} p(s\to s',1,\pi)\left[\phi(s') + \gamma \sum_{s''} p(s'\to s'',1,\pi)\nabla_\theta V^{\pi_\theta}(s'')\right]$$

$$= \phi(s) + \gamma \sum_{s'} p(s\to s',1,\pi)\,\phi(s') + \gamma^2 \sum_{s''} \left[\overbrace{\sum_{s'} p(s\to s',1,\pi)\,p(s'\to s'',1,\pi)}^{p(s\to s'',2,\pi)}\right] \nabla_\theta V^{\pi_\theta}(s'')$$

$$= \phi(s) + \gamma \sum_{s'} p(s\to s',1,\pi)\,\phi(s') + \gamma^2 \sum_{s''} p(s\to s'',2,\pi)\nabla_\theta V^{\pi_\theta}(s'')$$

$$= \phi(s) + \gamma \sum_{s'} p(s\to s',1,\pi)\,\phi(s') + \gamma^2 \sum_{s''} p(s\to s'',2,\pi)\phi(s'') + \cdots$$

$$= \sum_{k=0}^{\infty}\sum_x \gamma^{\,k} p(s\to x,k,\pi)\phi(x)\,, \qquad p(s\to x,k,\pi)\text{: prob. of state after } k \text{ steps from } s \text{ under } \pi$$

A4. per step view of $\nabla_\theta J(\theta)$ [proof – cont.]

$$\nabla_\theta J_0(\theta) = \sum_{s_0 \in \mathcal{S}} p_0(s_0) \nabla_\theta V^{\pi_\theta}(s) = \sum_{s_0} \sum_{s} \sum_{t=0}^{\infty} \gamma^t \, p_0(s_0) p(s_0 \to s, t, \pi) \phi(s)$$

$$= \sum_{s} \left[\sum_{s_0} \sum_{t=0}^{\infty} \gamma^t \, p_0(s_0) p(s_0 \to s, t, \pi) \right] \phi(s)$$

$$= \sum_{s} \rho^{\pi_\theta}(s) \phi(s), \qquad \rho^{\pi_\theta}(s) = \sum_{s_0} \sum_{t=0}^{\infty} \gamma^t \, p_0(s_0) p(s_0 \to x, t, \pi) : \text{discounted improper stationary dist.}$$

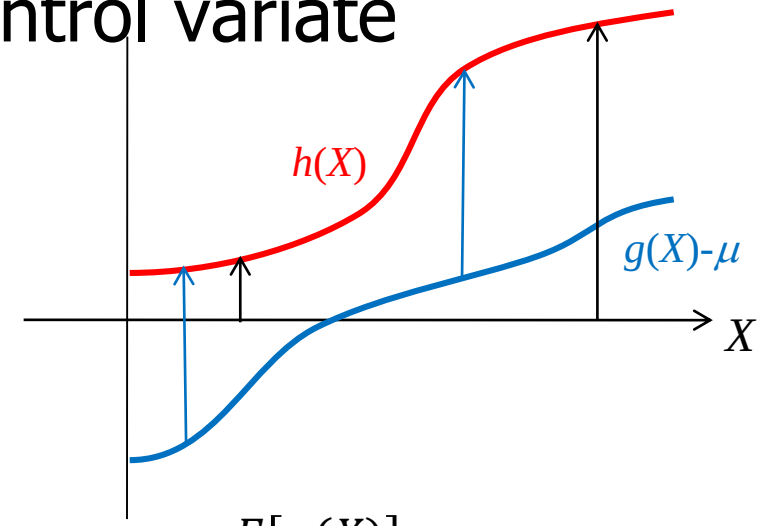
If we define $d^{\pi_\theta}(s) = \dfrac{\rho^{\pi_\theta}(s)}{\sum_{x \in \mathcal{S}} \rho^{\pi_\theta}(x)}$: discounted stationary dist. (proper)

$$\nabla_\theta J_0(\theta) \propto \mathbb{E}_{s \sim d^{\pi_\theta}}[\phi(s)] = \mathbb{E}_{s \sim d^{\pi_\theta}} \left[\sum_a \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \right] = \mathbb{E}_{s \sim d^{\pi_\theta}} \left[\sum_a \frac{\pi_\theta(a|s)}{\pi_\theta(a|s)} \nabla_\theta \pi_\theta(a|s) Q^{\pi_\theta}(s,a) \right]$$

$$= \mathbb{E}_{s \sim d^{\pi_\theta}} \left[\sum_a \pi_\theta(a|s) \frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} Q^{\pi_\theta}(s,a) \right] = \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta(.|s)} \left[\frac{\nabla_\theta \pi_\theta(a|s)}{\pi_\theta(a|s)} Q^{\pi_\theta}(s,a) \right]$$

$$= \mathbb{E}_{s \sim d^{\pi_\theta}, a \sim \pi_\theta(.|s)} [\nabla_\theta \log \pi_\theta(a|s) \, Q^{\pi_\theta}(s,a)] \quad \text{QED}$$

A5. Variance Reduction w/ Control variate



Let $A = E_X[h(X)]$ for an r.v. X

[Crude MC estimator] : $\hat{A} = \frac{1}{n}\sum_{i=1}^{n} h(x_i)$

[Control variate estimator]

Consider a control variate $g(X)$ with known mean $\mu = E[g(X)]$

Let $D = h(X) - b(g(X) - \mu)$
$E[D] = E[h(X)] - b(E[g(X)] - \mu) = E[h(X)] = A$

$\hat{A}_{CV} = \frac{1}{n}\sum_{i=1}^{n} D_i$, where $D_i = h(x_i) - b(g(x_i) - \mu)$

$Var[D_i] = Var[h(x_i)] + b^2 Var[g(x_i)] - 2bCov[h(x_i), g(x_i)]$
$Var[D_i] < Var[h(x_i)]$ if $b^2 Var[g(x_i)] < 2bCov[h(x_i), g(x_i)]$
$Var[D_i]$ is minimized when we choose $b = \frac{Cov[h(x_i), g(x_i)]}{Var[g(x_i)]}$

$h(\tau) = \nabla_\theta \log p(\tau;\theta) R(\tau)$

$g(\tau) = \nabla_\theta \log p(\tau;\theta)$, $E[g(\tau)] = 0$

$Var[g(\tau)] = E[g(\tau)^2] = E[\nabla_\theta \log p(\tau;\theta)^2]$

$Cov[h(\tau), g(\tau)] = E[h(\tau) g(\tau)]$
$= E[\nabla_\theta \log p(\tau;\theta)^2 R(\tau)]$

$b^* = \frac{\sum_i score(\tau^{(i)})^2 R(\tau^{(i)})}{\sum_i score(\tau^{(i)})^2}$

$score(\tau^{(i)}) = \nabla_\theta \log p(\tau^{(i)};\theta)$